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COURSE NAME: Software Engineering	COURSE NUMBER: CS3151	SECTION: 1, 2, 3

Lab Objectives:

- Identify target users for the software project
- Develop realistic user personas
- Map personas to software requirements
- Apply user-centered design thinking

Part 1: Identify User Groups

Task: List all potential user groups of your system.

Deliverable:

User Group	Role	Frequency of Use	Technical Level
Active Students / Fitness Users	Use BodyBalance AI to check their physical condition, receive activity or recovery recommendations, and track their progress.	Daily / Several times per week	Medium
Fitness Coaches	Use the system to review client fitness and recovery information and adjust training recommendations.	Daily	Medium
Healthcare Professionals	Review user-authorized health information and recovery history to support their professional assessment.	Occasional	High

Part 2: Persona Creation

Create 2–3 personas using the following template:



1st Persona Name:	Sara Ahmed (Active Student / Fitness User)
Age:	22
Education:	University Student
Role:	University student and regular gym user
Goals:	• Stay physically active and healthy. • Improve her fitness. • Avoid making minor injuries worse. • Know when she should exercise or recover. • Track her physical progress.
Motivations:	• Wants personalized recommendations instead of general workout advice. • Wants to understand the possible reasons behind recurring pain or discomfort. • Wants to exercise safely while maintaining an active lifestyle.
Pain Points:	• Sometimes experiences muscle pain after workouts. • Is unsure whether she should continue exercising or rest. • Does not know when pain requires professional attention. • Existing fitness apps do not always consider her current physical condition.
Technical Skill Level:	Medium–High
Usage Context:	Sara uses BodyBalance AI after workouts or when she experiences pain, fatigue, stress, or physical discomfort. She completes a short check-in and can upload an image of the affected area or her posture for BodyScope AI analysis.

2nd Persona Name:	Omar khalid (Fitness Coach / Personal Trainer)
Age:	29
Education:	Bachelor's Degree in Sports Science
Role:	Fitness Coach / Personal Trainer



Goals:	• Provide safer and more personalized training. • Understand his clients' physical condition. • Track client recovery. • Reduce training-related problems and unnecessary injuries. • Adjust exercises based on client feedback.
Motivations:	• Wants better information about clients before recommending exercises. • Wants to personalize workouts according to each client's condition. • Wants to recognize when a client may need professional evaluation.
Pain Points:	• Clients may not always describe their pain accurately. • Has limited time to assess every client's physical condition. • It can be difficult to track changes in clients' symptoms over time. • He cannot diagnose medical conditions.
Technical Skill Level:	Medium
Usage Context:	Omar uses BodyBalance AI with his clients before or after training. With the client's permission, he reviews relevant fitness and recovery information and uses it to adjust the client's training plan.

3rd Persona Name:	Ahmed Hassan (Healthcare Professional – Sports Medicine / Physiotherapy)
Age:	38
Education:	Doctor of Medicine
Role:	Sports Medicine / Physiotherapy Professional
Goals:	• Receive organized information about a patient's symptoms and activity. • Understand the patient's condition and history before an assessment. • Monitor recovery progress. • Identify patients who may need further evaluation.



Motivations:	• Wants organized information about symptoms and physical activity. • Wants to understand changes in the patient's condition over time. • Wants technology to support clinical assessment without replacing professional judgment.
Pain Points:	• Patients may forget details about when symptoms started. • Information about previous symptoms and physical activity may be incomplete. • Reviewing long-term recovery information manually can be difficult.
Technical Skill Level:	High
Usage Context:	Dr. Ahmed reviews user-authorized symptom, activity, and recovery information through BodyBalance AI. The information supports his professional assessment but does not replace medical diagnosis or treatment.

Part 3: Persona to Requirements Mapping

Map each persona to required system features.

Persona	Feature Needed	Priority
Sara Ahmed (Active Student / Fitness User)	Daily physical and mental check-in	High
Sara Ahmed (Active Student / Fitness User)	Pain Locator	High
Sara Ahmed (Active Student / Fitness User)	BodyScope AI image analysis	High
Sara Ahmed (Active Student / Fitness User)	Personalized activity recommendation	High



Persona	Feature Needed	Priority
Sara Ahmed (Active Student / Fitness User)	Personalized recovery plan	High
Sara Ahmed (Active Student / Fitness User)	Professional referral	High
Sara Ahmed (Active Student / Fitness User)	Progress tracking	Medium
Omar Khalid (Fitness Coach / Personal Trainer)	Client profile	High
Omar Khalid (Fitness Coach / Personal Trainer)	View client activity and recovery information	High
Omar Khalid (Fitness Coach / Personal Trainer)	Personalized training recommendations	High
Omar Khalid (Fitness Coach / Personal Trainer)	Recovery monitoring	Medium
Omar Khalid (Fitness Coach / Personal Trainer)	Physical condition alerts	High
Ahmed Hassan (Healthcare Professional - Sports Medicine / Physiotherapy)	Patient-authorized health summary	High
Ahmed Hassan (Healthcare Professional - Sports Medicine / Physiotherapy)	Symptom and activity history	High
Ahmed Hassan (Healthcare Professional - Sports Medicine / Physiotherapy)	Recovery timeline	Medium
Ahmed Hassan (Healthcare Professional - Sports Medicine / Physiotherapy)	Professional referral information	Medium
Ahmed Hassan (Healthcare Professional - Sports Medicine / Physiotherapy)	Clinical decision-support information	High

Part 4: Scenarios & User Stories

Part 4: Complete System Scenarios & User Stories

The following scenarios describe how different users may interact with BodyBalance AI from the moment they enter the system until they complete their session. These scenarios include the main personas as well as other possible users and situations that may be supported as the system develops.



A. 4.1 User Registration Scenario

A new user opens BodyBalance AI for the first time and chooses to create an account. The system asks for basic information such as name, age, email, and password. The user reviews and accepts the privacy and consent information before creating the account. After successful registration, the user is directed to the initial profile setup.

User Story:

As a new user, I want to create an account so that I can securely use BodyBalance AI and save my information and progress.

B. 4.2 Login and Password Recovery Scenario

A registered user opens BodyBalance AI and logs in using their account information. If the user forgets the password, they can request a password reset and follow the verification process before accessing the account again.

User Story:

As a registered user, I want to securely log in or recover my password so that I can access my account and previous information.

C. 4.3 Initial Profile Setup Scenario

After registration, the system asks the user to complete a basic profile. The user provides information such as age, activity level, fitness background, goals, and relevant physical limitations. This information helps the system personalize future recommendations.

User Story:

As a user, I want to create my personal profile so that the system can provide recommendations that are more relevant to me.

D. 4.4 Daily Physical and Mental Check-In Scenario

The user completes a short check-in about their current physical and mental condition. The system may ask about pain, fatigue, energy, stress, sleep, recovery, and recent physical activity. The information is saved and considered when providing recommendations.

User Story:

As a user, I want to report my current physical and mental condition so that the system can understand how I am feeling before giving recommendations.



E. 4.5 Pain Locator Scenario

A user experiencing pain selects the affected area using the Pain Locator. The system asks additional questions about the pain, such as when it started, its intensity, whether it followed an activity or injury, and whether it has changed over time.

User Story:

As a user experiencing pain, I want to identify the affected area and describe my symptoms so that the system can better understand my condition.

F. 4.6 BodyScope AI Image Analysis Scenario

When visual information may be useful, the user can upload an image of the affected area or their posture. Depending on the situation, the system may request a close-up image of the injury, a wider image showing the surrounding area, or a full-body/posture image. BodyScope AI analyzes available visible indicators and combines them with the user's reported symptoms and activity information.

The system clearly informs the user that the analysis does not replace professional medical diagnosis.

User Story:

As a user, I want to upload relevant images so that BodyScope AI can consider visible physical indicators together with the information I provide.

Scenarios for Different User Groups

G. 4.7 Active Student / Fitness User Scenario

An active student or fitness user experiences soreness, pain, fatigue, or discomfort before or after exercise. The user completes the check-in and may use the Pain Locator or BodyScope AI. The system reviews the available information and provides personalized activity or recovery guidance. Depending on the situation, the system may suggest continuing normal activity, modifying the workout, reducing intensity, resting, or seeking professional evaluation.

User Story:

As an active student or fitness user, I want recommendations based on my current physical condition so that I can exercise more safely.

H. 4.8 MMA / Combat Sports User Scenario

An MMA, boxing, or combat-sports user experiences bruising, muscle pain, joint discomfort, or another training-related concern after training or competition. The user reports what happened, identifies the affected area, and may upload relevant images. The system considers the injury information, recent activity,



and available visual indicators before providing appropriate recovery guidance or recommending professional evaluation when necessary.

User Story:

As a combat-sports user, I want to report and track training-related injuries so that I can better understand when to recover, modify training, or seek professional evaluation.

I. 4.9 First-Aid Scenario

A user opens BodyBalance AI after an injury occurs and selects the first-aid option. The system asks what happened, where the injury occurred, what symptoms are present, and whether any concerning warning signs exist. Based on the information provided, the system gives appropriate basic guidance within its supported scope.

If the information suggests that the situation may require urgent professional attention, the system prioritizes seeking emergency or professional care instead of continuing with normal recommendations.

User Story:

As a first-aid user, I want clear guidance after an injury so that I can understand the appropriate next step and recognize when professional help may be needed.

J. 4.10 Older Adult Scenario

An older adult uses BodyBalance AI because of pain, discomfort, mobility difficulty, or concerns related to physical activity. The system considers the user's age, reported limitations, symptoms, activity level, and available history before providing suitable guidance.

Recommendations are adjusted to avoid suggesting activities that may be inappropriate for the user's reported condition. If concerning symptoms are reported, professional evaluation is recommended.

User Story:

As an older adult, I want recommendations that consider my physical condition and limitations so that I can manage activity and recovery more safely.

K. 4.11 Posture and Movement Scenario

A user wants to understand their posture or possible movement-related concerns. The system requests an appropriate full-body or posture image and relevant information about discomfort or physical activity. BodyScope AI analyzes visible posture information and provides non-diagnostic observations and appropriate recommendations.



User Story:

As a user, I want to analyze my posture and movement-related concerns so that I can become more aware of possible physical patterns affecting my activity.

L. 4.12 Visible Injury or Bruising Scenario

A user notices bruising, swelling, or another visible physical change. The user uploads an image and answers questions about how the injury occurred, when it appeared, pain level, and related symptoms. The system organizes the information and provides guidance based on the available data.

If the reported information indicates that professional assessment may be necessary, the system recommends seeking appropriate care.

User Story:

As a user with a visible injury, I want to document and track the affected area so that I can monitor changes and know when professional evaluation may be appropriate.

M. 4.13 Possible Violence or Abuse-Related Injury Scenario

A user may use the system to document an injury that could be related to violence or abuse. The system handles the information privately and avoids requiring the user to identify the cause if they do not feel safe doing so. The user may record symptoms and visible injuries and, when appropriate, receive information encouraging professional medical care or relevant support resources.

The system should prioritize privacy, user safety, and discreet access to information.

User Story:

As a user in a sensitive situation, I want to privately document physical injuries and access appropriate support information so that I can make informed decisions about seeking help.

Recommendation and Recovery Scenarios

N. 4.14 Personalized Activity Recommendation Scenario

After collecting information from the user's profile, check-in, symptoms, recent activity, and available BodyScope analysis, the system generates a personalized activity recommendation. Depending on the available information, the recommendation may suggest maintaining, reducing, modifying, or temporarily stopping certain physical activities.

User Story:

As a user, I want activity recommendations based on my current condition so that I can make safer decisions about exercise and physical activity.



O. 4.15 Personalized Recovery Plan Scenario

When recovery is appropriate, the system provides a personalized recovery plan based on the user's reported condition and activity. The user can follow the recommendations and return later to report changes in symptoms.

User Story:

As a user, I want a personalized recovery plan so that I can follow appropriate steps and monitor my progress.

P. 4.16 Follow-Up and Progress Tracking Scenario

After a previous check-in or injury report, the system asks the user for a follow-up. The user reports whether pain, discomfort, mobility, or other symptoms have improved, remained the same, or become worse.

The system records these changes and creates a progress history that the user can review over time.

User Story:

As a user, I want to track changes in my condition over time so that I can understand whether I am improving or may need additional help.

Q. 4.17 Concerning Symptoms and Escalation Scenario

If a user's answers or reported condition contain concerning indicators, the system does not continue with normal fitness recommendations as if the situation were routine. Instead, it informs the user that professional or urgent medical evaluation may be appropriate.

User Story:

As a user with concerning symptoms, I want the system to recognize when normal fitness or recovery guidance may not be appropriate so that I know when to seek professional help.

R. 4.18 Professional Referral Scenario

When professional evaluation is recommended, BodyBalance AI provides the user with a summary of relevant information collected during the session, such as symptoms, affected area, activity history, recovery progress, and available BodyScope observations.

This information may help the user communicate their situation more clearly when seeking professional care.



User Story:

As a user who may need professional evaluation, I want an organized summary of my relevant information so that I can communicate my situation more clearly to a healthcare professional.

Professional User Scenarios

S. 4.19 Fitness Coach / Personal Trainer Scenario

With the client's permission, a fitness coach can access relevant fitness, activity, and recovery information. The coach can review patterns such as recurring discomfort after specific exercises and use the information to adjust the client's training plan.

The coach does not receive access to information that the user has not authorized.

User Story:

As a fitness coach, I want to review authorized client activity and recovery information so that I can personalize training more safely.

T. 4.20 Healthcare Professional Scenario

With the user's permission, a healthcare professional can review an organized history of reported symptoms, physical activity, recovery progress, and previous BodyScope observations. The information supports the professional's assessment but does not replace their clinical judgment, diagnosis, or treatment.

User Story:

As a healthcare professional, I want to review authorized patient information and recovery history so that I have organized information available during my professional assessment.

Privacy and Account Management Scenarios

U. 4.21 Permission and Information-Sharing Scenario

When a user wants to share information with a fitness coach or healthcare professional, the system asks for permission before providing access. The user can choose what information is shared and can later change or remove access.

User Story:

As a user, I want control over who can access my information so that my personal information remains private.



V. 4.22 History Review Scenario

A user can open their history to review previous check-ins, reported symptoms, BodyScope observations, recommendations, recovery plans, and progress information.

User Story:

As a user, I want to review my previous information so that I can understand changes in my physical condition over time.

W. 4.23 Profile Management Scenario

The user can update relevant profile information, such as activity level, goals, or other editable information. The system uses updated information when generating future recommendations.

User Story:

As a user, I want to update my profile so that the system uses accurate information when personalizing my experience.

X. 4.24 Logout and Session Security Scenario

When the user finishes using BodyBalance AI, they can securely log out. The system protects saved information and requires appropriate authentication before the account can be accessed again.

User Story:

As a user, I want to securely log out of my account so that my personal information remains protected.

Assessment Criteria:

Criterion	Marks
Persona realism & clarity	30%
Alignment with project idea	30%
Requirements mapping	20%
Scenarios & user stories	20%